

The evolution of conferencing

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In the last few years we have seen the explosive growth of the personal computer in the business and home environment, together with the networking of these computers. This has presented new opportunities to enhance the traditional audio- and video-conferencing environments that have been around for the last decade.

1. A brief history

Visual communication has a long history in the UK with the first public service nearly 25 years ago called 'Confravision', which linked London, Manchester, Birmingham, Bristol and Glasgow [1]. Soon after this advances were made in video coding techniques in which it was possible to compress the video signal and transmit it over digital circuits. This work was ground breaking not just because 100 to 1 compression was achieved on the video pictures, but also because for the first time companies collaborated together to define the standards, thus recognising that people ultimately needed to be able to communicate outside closed user groups [2].

The first standard was the European H.120 standard for the transmission of audio and video over 2 Mbit/s links. In the mid-1980s it was recognised that it was now technically feasible to utilise the integrated services digital network (ISDN) for the transmission of audio and video. The ISDN provided a digital connection end to end, image compression techniques showed acceptable picture quality at 128 kbit/s and close to VHS quality at 2 Mbit/s, and field programmable devices were appearing on the market which meant development costs were coming down. Therefore, at that time, the International Telecommunications Union (ITU), formerly the CCITT, began work on the H.320 standard which is the cornerstone of videoconferencing today [3].

The first commercial implementation of H.320 was developed at BT and was known as the VC2100 codec. This codec was able to operate at all data rates from 64 kbit/s to 2 Mbit/s implementing many of the optional operating modes of H.320 and became a benchmark for the industry.

These H.320 systems were still room based in the early 1990s, but the bit rates at which they operated came down from 2 Mbit/s to 384 kbit/s as users became aware that the video quality from the H.261 video codec was perfectly acceptable at these lower bit rates. The cost of H.320 room-

based systems started to come down in price as very large scale integration (VLSI) became available for the computationally intensive parts of the system. With such VLSI the physical size of codecs was reducing all the time and BT embarked on a personal computer (PC) based H.320 system. This led to the familiar PC videophone, the VC8000, which was launched in January 1994.

The VC8000 revolutionised the H.320 conferencing market; H.320 became more affordable with an entry price of £3500 in 1994 but more importantly it harnessed the power of the PC to bring data collaboration. Users of PC videophones were able to pass files from one PC to another PC during the call, were able to view documents together and annotate them, and were able to share applications such that a person who did not have the application on their PC could view the remote application and control it. This was all achievable through the use of another set of important standards from the ITU designated T.120.

The next revolution was the astonishing pace with which PC performance was growing year on year. In the space of the three years that the VC8000 has been on the market, the PC has moved from the 386 to the 486, through the Pentium and on to the Pentium Pro, and now the Pentium II. The central processor unit (CPU) performance has been doubling almost every 18 months. Such processor power soon made it possible to harness the CPU to perform the codec functions of H.320.

It was also the era in which the growth of networked computers, i.e. the Internet but more importantly intranets, exploded. With new protocols such as RSVP (the resource reservation set-up protocol), it was soon going to be possible to use local area networks (LANs) as a means of transporting real-time audio and video traffic. Hence the ITU set about defining a standard by which audio/video-conferencing could take place over LANs — this became known as H.323. H.323 has become an important standard,

with significant industry backing from heavy-weights such as Intel and Microsoft.

Image and voice compression techniques have also not stood still over time (see Whybray et al [4]). In 1990 H.261 was state of the art in video compression algorithms for low bit rates delivering 10 to 15 pictures per second at ISDN BRI rates with 30 pictures per second at 384 kbit/s plus. The coding was a motion-compensated interframe prediction loop of which the prediction error was subsequently transformed using the discrete cosine transform (DCT), quantised and variable length coded. The VC2100 was the first codec to implement motion compensation in real time [5]. It used a 'full search' algorithm which meant that for every 16×16 pixel video block there was an exhaustive search of 961 possible positions in the previous coded frame in which the best match could be found. The computational power required was equivalent to 5800 million operations per second, leading to the largest application-specific integrated circuit (ASIC) to be designed in Europe at the time.

Improvements have subsequently been made to the motion compensation, interpolated frames have been added to improve frame rate at low bit rates, and a number of other improvements to the coding which have brought about an approximate 2:1 improvement in performance overall; this has been incorporated into the H.263 standard. The audio has also been improved with the latest generation of codecs transmitting near toll-quality speech at 6.3 kbit/s (G.723.1) instead of 16 kbit/s (G.728) before.

Such improvements have meant it is now possible to videoconference over analogue telephone lines using the latest generation of modems (V.34) and the improved audio and video codecs mentioned above. The ITU have formally approved a standard for this (H.324) and there are now products on the market.

2. Benefits

The ability to communicate over distance brings about major benefits to companies such as the speed of decision making by reducing the need for physical meetings and hence the travel time associated with them. Costs can be reduced and key personnel can make better use of their time. This has been the power of the global voice network today, keeping people and businesses in touch. There are, however, many situations where voice-only communication is insufficient and there is a need to be able to see each other, for instance in contract negotiations body language is just as important as the spoken word. In other situations there may also be the need to share information such as when a mortgage expert assists an applicant to fill in mortgage forms remotely.

The benefits have been well understood over time but the videoconferencing market is still niche and the expected uptake of desktop conferencing has not yet materialised.

Why should this be? The answer is that to realise the benefits detailed above places quality thresholds on the audio-, video- and data-conferencing in the particular application for which it is used.

2.1 The important factors

In a videoconference the video transmission quality is determined by the resolution and the number of pictures per second. We as users compare the quality against the images we view daily from our television set or video cassette recorder (VCR) at home. With limited bandwidth from the public switched telephone network (PSTN), ISDN or LAN, it is not feasible to deliver such quality end to end. For a typical head and shoulders scene, pictures are delivered over ISDN2 (128 kbit/s) with a spatial resolution approximately a quarter that of VHS and at a picture rate approaching 15 pictures/sec; over PSTN (28.8 kbit/s) the picture rate would be approximately 8 pictures/sec for the same image resolution. Higher bandwidths would be required to reach close to VHS quality, 384 kbit/s and above, with the obvious added cost penalty of increased call charges. This means that applications have to pick carefully the appropriate image quality.

As important is audio quality, which includes lip synchronisation and end-to-end delay. If video quality is degraded, the call is not necessarily interrupted; on the other hand, without stable and acceptable audio quality, a video-conference call can be a very irritating experience! The longer the call, the greater the quality required.

For a data conference, the most important aspect is the integrity of the information exchanged. For example, with a multiple file transfer it is critical that the file be sent without error to each participant. Speed of transmission is also important — annotations on a shared whiteboard need to be immediately displayed on each person's screen as they happen. Here too, delays or interference detract from the efficiency of the conference.

Today, choosing the right videoconference or data conference solution for professional usage means determining which technology and network delivers the optimum trade-off between reliability, speed and cost.

3. Today's conferencing

Businesses are supported with a full range of conferencing solutions today, ranging from custom studios on dedicated corporate networks to desktop devices either incorporated into personal computers or as stand-alone. All BT's ISDN videoconferencing solutions are based on ITU's H.320 Recommendation which means that the customer's investment is protected, there is global coverage and support, and interworking is ensured.

All the solutions provide a rich environment for communications. Conference room systems offer high quality audio and video at bit rates from 384 kbit/s to 2 Mbit/s plus control of a number of peripherals, both at the local and far ends of the link. The desktop PC solution has the added advantage of providing sophisticated data-sharing capabilities across the link alongside the audio and video, though this is soon to come to the room systems as well.

3.1 Desktop systems

This is a system which offers a range of data conference features, such as a shared 'whiteboard', application sharing or multiple file transfer, together with audio- and video-conferencing on a PC. These generic data features can be tailored to suit particular applications.

Whiteboard

The whiteboard is an on-screen workspace viewed by everyone in the conference and is similar to the whiteboard you find in a meeting room. Images can be loaded on to this screen, such as presentations or drawings from other applications, which can subsequently be annotated by all in the conference. The original material that is imported into the whiteboard is not changed, it simply forms one of the presentation layers; each conference end-point is allocated a layer in which their annotations are drawn.

Application sharing

Application sharing is subtly different to whiteboarding, as described above, in that the data that is shared is permanently altered by any annotations carried out. Only one conference participant needs to have the application and work material, the other conference participants are given the ability to share the PC resources on which the application is running. The resources shared are the keyboard strokes, mouse movements and graphics. This technology is platform-specific at present as it involves intercepting commands and information at the operating system level, hence its restriction to common operating systems in a conference.

Multiple file transfer

Multiple file transfer enables a file to be sent simultaneously to two or more participants at remote sites. For instance, a spreadsheet that has been discussed and changed using application sharing, as described above, in a conference can be copied to all or a selection of participants in the conference at the end.

3.2 Group systems

Group systems are more scalable offering a range of options in terms of available data rates and the numbers of cameras and screens. The conference user has much greater control of the conference with the ability to control both near and far ends. Group systems also harness the flexibility of a PC for such things as providing the data-conferencing applications and the ability to control the conference through a graphical user interface.

3.3 Multipoint conferences

The discussion so far has centred around systems that are primarily designed to operate on a point-to-point basis, facilitating communications between two end-users or a user and a network service. Increasingly there is a need to communicate between three or more locations, which is supported through a device known as a multipoint control unit (MCU) [6].

Multiple sites are supported by a star network which joins each location to a central node. At this hub, information can be processed, selected and forwarded to all other locations in the star (see Fig 1). Such a configuration does not require extra equipment at the location end points and can be expanded by cascading MCUs together to form a dumb-bell type arrangement, as shown in Fig 2. Such an arrangement can be useful to concentrate traffic from several sites together before transmission across international links for instance.

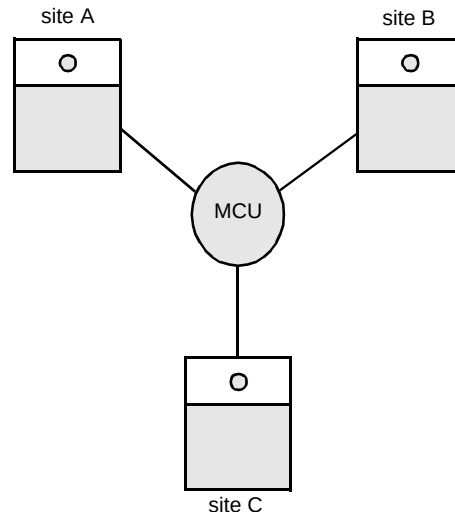


Fig 1 Typical star network with three locations.

Switched multipoint

These are conferences in which the video information is selected by some means and distributed to all other connected locations. The video can be switched automatically such as by voice activation or it could be manually switched under the control of a chairperson. The

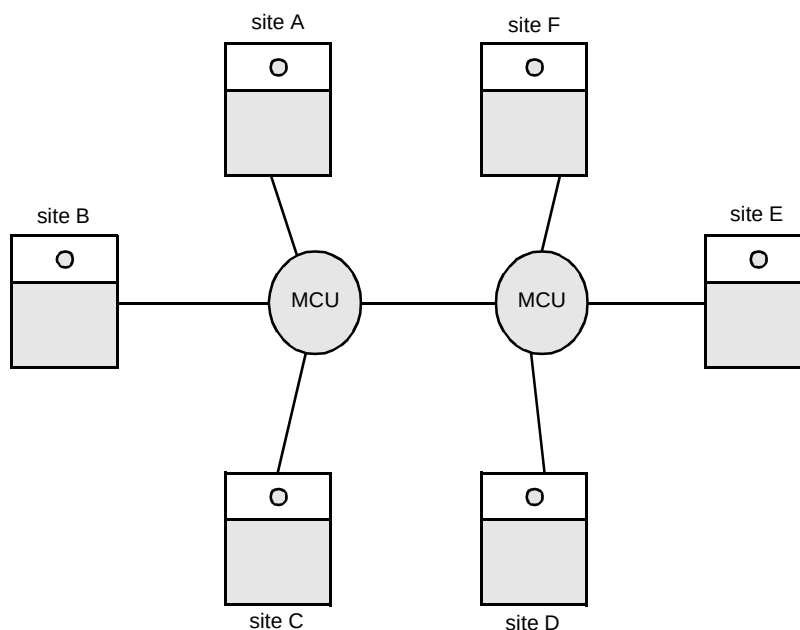


Fig 2 Typical dumb-bell network with six locations.

audio from each site is mixed together centrally and distributed to all sites.

This is a non-ideal environment for multipoint conferences because each location only sees the location that is speaking, and as a consequence visual cues of other participants are lost in the discussion. If the conversation continually moves between locations then the video switches frequently. The video could also switch on a false trigger, such as a cough or door closing, unless under chairperson control. A more natural environment would be if all locations could see each other during the conference, this is known as continuous presence multipoint.

Continuous presence

It is now possible for MCUs to process the video from up to four locations and to distribute to all others, providing the ability to hold a continuous presence conference with up to four locations. This significantly improves the quality of a multipoint videoconference — the screen is not now continually changing in front of you and the visual cues from others in the conference are not now lost.

3.4 Conferencing services

Audioconferencing is the backbone of any conferencing service provider as depicted in Fig 3 and probably accounts for 75% of the market today. It is a low-cost solution with no barriers to entry, access is via the ubiquitous telephone, and a high quality of service can be offered.

Videoconferencing has a much higher cost of entry and is more difficult to use and hence has a smaller share of the overall conferencing market. These types of call require

significantly more intervention from the service provider to ensure the calls are successful due in a large part to the immaturity of the standards and equipment in this market.

Between pure audioconferencing and videoconferencing lies audiographic conferencing. Audiographic conferencing is still in its infancy with an early set of standards, T.120/T.130, that need to 'bed down'. The cost of entry will be low as it is a software application run on a PC and, once there is a population of conference-'enabled' PCs, this end of the market will grow. Two connections are required, one for voice and one for data. For a teleworker this could be two PSTN lines or one ISDN2 line. For someone working from their office and whose PC is already connected to the corporate LAN, their existing LAN connection could be used for the data and their phone for the voice.

There are also a growing number of one-line solutions — voice/data modems (V.70), the use of H.324 or H.323 clients, and voice/data over Internet protocol (IP). With V.70 and H.324 having only appeared in late 1996 and with a number of voice-over-IP implementations, it is difficult to determine which solution will prevail at the end of the day.

4. VCnet

One of the largest managed videoconferencing networks is VCnet within BT, with over 55 sites in the UK (see Fig 4) and other sites at country offices around the globe. Each studio room has been specially created to conform with specialist guide-lines on size, lighting, furnishing, air-conditioning and sound-proofing. The criteria for location of facilities is based on having a room within 30 minutes travelling time of 80% of managers.

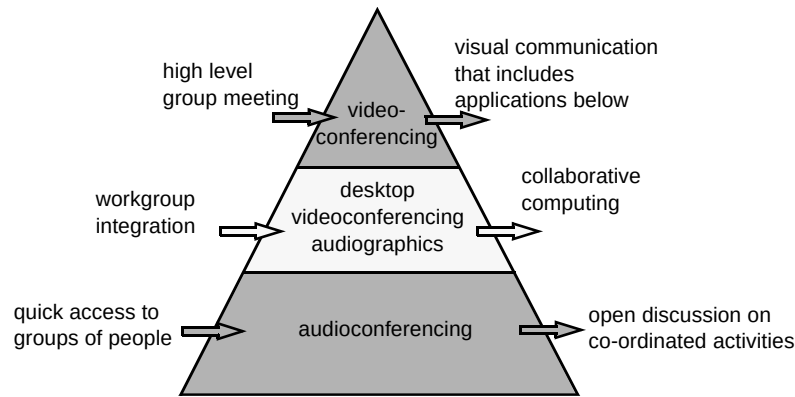


Fig 3 Conferencing needs and tools.



Fig 4 Map of VCnet sites in the UK.

The equipment is all H.320 standard-compliant and at each site there is a standard build for each facility of a VC5000 double unit accommodating up to six participants per site and an ELMO graphics unit or a display station for document transmission. Some sites are equipped with a VCR and a PC interface. Soon the VS3 will be introduced into the studios which will bring the added benefit of data collaboration.

All sites are operated over the IBTN, a Timeplex Link network, and use 384 kbit/s channels connected locally by direct-cabled RS449 interface, or a 384 kbit/s KiloStream circuit. Some sites are equipped with ISDN6 dialling capability to allow direct dialling to sites outside BT.

The network is centrally managed out of Pontefract and there are five VC4100 multipoint control units (MCUs) each of 8-port capacity within the network, and a new VC4600 MCU is being brought into service, allowing far greater ISDN 2B and 6B interconnectivity.

All conferences are pre-booked via a telephone reservation desk and the IBTN network is configured dynamically to set up the relevant 384 kbit/s connections. All connections are established prior to the conference start time ensuring that when participants arrive at the studios the connections are already up and running and the meeting can begin at once.

5. On the horizon

5.1 Internet conferencing

The Internet of the mid-1990s suffers from the chronic shortage of bandwidth and the total lack of quality-of-service levels which makes it incompatible with intensive professional use for teleconferencing applications. However, the Internet is here to stay and, with continued investment and the deployment of protocols in the network that support quality of service (QOS), the Internet will offer significant flexibility and an unprecedented platform for creativity for conferencing in the future.

The first Internet conferencing applications, based on the H.323 standard, are currently available for trials. They integrate standard collaborative working applications such as whiteboards and application sharing, utilising T.120, as well as Internet telephony services. Significantly they are all PC-based solutions and the majority are software-only solutions.

Given the levels of investment being sunk into Internet solutions by both IT and telecommunications companies, it is likely that within 18 months the Internet will become an attractive alternative for data conferences and that Internet-based videoconferencing will be feasible within two years. Not only will PC processing power be even greater, codec technologies will also have been optimised, and bandwidth-on-demand protocols for the Internet will have been standardised and deployed, guaranteeing users a certain quality of service.

Where the real revolution is happening is in corporate networks, or intranets as they have become known, which are now ubiquitous in large businesses. H.323 offers large and small corporations the ability to route telephony, data and real-time multimedia services to the desktop over a single connection — the office LAN. It allows provision of service from basic telephony right through to sophisticated computer/telephony integrated call management. With corporations being able to reserve IP network bandwidth, H.323 will be the inter-site communication connection of choice.

H.323 has been developed to extend the ITU's H.320 set of videoconferencing standards to operate over non-guaranteed quality of service LANs (such as Ethernet) and the Internet.

H.323 systems will interwork with H.320 systems on the ISDN and H.324 systems on the PSTN by way of gateways (see Fig 5). In addition to the terminals that sit on the LAN side, there are also gatekeepers. These are used for bandwidth-reservation (or perhaps more accurately bandwidth-restriction) and address-translation services. Thus terminals, gateways, and gatekeepers are the three main functional units described in H.323. It should be noted, however, that, in practice, the gatekeeper functionality might be implemented as part of a particular vendor's gateway unit, or, indeed, be implemented in one of the terminals.

5.2 Conference control

The conferencing environment will become richer in time; the T.120 protocols, which provide the infrastructure for data conferencing, are about to be enhanced with a further set of protocols that allow the manipulation of audio and video media streams in a conference. This set of protocols is currently being defined in the ITU and is known as T.130. The services provide support for user controlled distribution of real-time streams and automated stream processing and distribution services such as video switching, audio mixing, and continuous presence. Other services include stream identification, floor control and 'on-air' indications. In addition, peer-to-peer functions such as source selection, privacy and remote device control are supported. A forerunner to this is a BT development called Passepartout [7], which has set out to develop a richer environment for audio conferencing by linking audio and data conferencing together in order to generate and prove some of the more complex protocols mentioned above.

The implementation consists of an audio bridge which can be controlled through an applications programme interface (API), in this case the minor applications platform (MAP), a data bridge which connects to the MAP's API and to conference participants' PCs through the corporate intranet or a direct-dial connection, with a PC client providing a conference control graphical user interface.

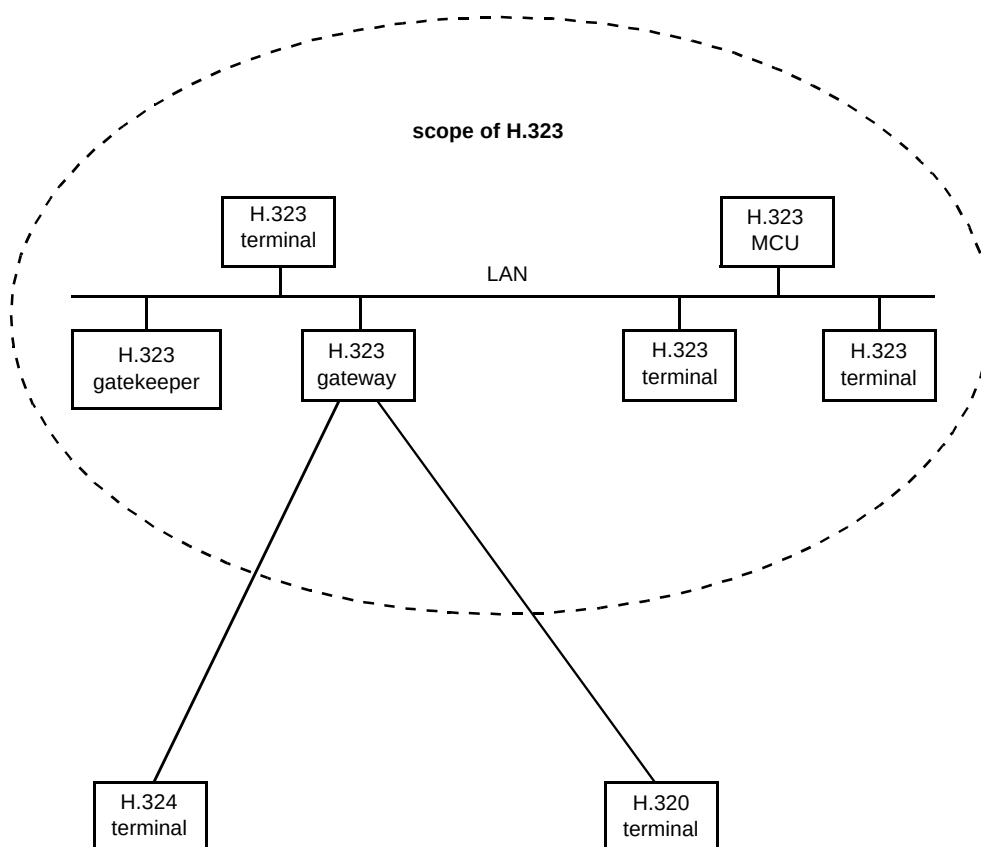


Fig 5 Scope of H.323.

The first implementations of Passepertout will provide the following services:

- dial-out audio,
- graphical indication of who is in the conference,
- graphical chairperson's control for mute and talk,
- dataconferencing application support for file transfer, whiteboard and application sharing,
- rapid conference set-up through e-mail.

5.3 Broadband

In time the availability of bandwidth will increase, the current restrictions in the local access loop will be overcome to bring bit rates in excess of 2 Mbit/s to homes through the use of technologies such as ADSL and VDSL [8]. In the corporate LAN, improved switching and segmenting the networks will bring 100 Mbit/s plus to the desktop.

Increased bandwidth offers the opportunity to deliver high-quality conferencing with TV-quality images and stereo audio, making the user experience of conferencing much closer to the real thing.

6. Conclusions

Teleconferencing is entering a new era in which new conferencing protocols are able to harness the flexibility of PCs and computer networks to bring a richer environment to users. The advent of conference-enabled personal computers by the year 2000 will stimulate a whole host of new services, some of which have been touched on in this paper.

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He joined BT Laboratories in 1986 working initially on the design of a motion estimator chip for an H.320 videoconferencing codec. He then went on to study variable bit rate video coding techniques and represented BT at ITU standards meetings. In 1991 he became a member of the ISO-MPEG group developing video coding standards for storage and retrieval, with a specific interest in layered coding approaches. In 1993 he joined the videotelephony services group involved (in collaboration with IBM, ICL and Olivetti) in the launch of the PC Videophone. He is currently working in BT's On-Line and Multimedia sector responsible for developments on broadcast and conferencing products and services. He represents BT on the board of the International Multimedia Teleconferencing Consortium.